

ME 304: Power Plant Engineering / Principles of Energy Conversion

Course Code	ME 304
Course Name	Power Plant Engineering / Principles of Energy Conversion
L-T-P-C	3-0-2-4
Prerequisites	Thermodynamics, Fluid Mechanics
Mutual Exclusion	ME 356 (ME 304 is equivalent to ME 356).
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Introduction** Energy sources. Electricity generation, Major issues nationally, internationally.
- **Economics Electric power systems** The grid; Supply - generating plants, availability, demand-supply mismatch, grid collapse. Grid regulation
- **Revision of thermodynamics basics** Basic thermodynamics, Rankine cycle and its variants
- **Thermal Energy Generation:**
- **Fossil Fuelled Power Plants** system analysis, turbine generators, condenser and cooling water systems, condensate system, feed water system, boilers, furnace, fuel and systems, boiler auxiliary systems
- **Gas Turbine Power Plants** Gas turbine fundamentals, gas turbine engineering, combined cycle analysis
- **IC Engine Power Plant** IC Engine-based systems Solar Thermal Power Plants: Concepts and basics, System engineering
- **Earth-based thermal power plants** Geothermal, Ocean Thermal
- Biomass, waste sources-based Energy
- **Nuclear power** Basics, Nuclear reactions basics, Reactor engineering, Light water reactors, Heavy water reactors, Fast Reactors, Other types of reactors. Fusion. Issues & concerns
- **Fluid Flow Based Power Generation:**
- **Hydroelectric plants** Introduction, Hydro-turbine basics, Hydro-power plant engineering
- **Wind electric generators** Windmill basics, Windmill engineering
- **Ocean-based power plants** Waves and tidal power
- **Direct Conversion:**
- **Solar photovoltaic** SPV fundamentals, SPV engineering
- **Fuel cells** Fuel cells fundamentals, Fuel Cell types, Engineered systems
- **MHD generation** MHD generation

Text and Reference Books

1. Clapp, R. M., (Ed.), **Modern Power Station Practice: Boilers and Ancillary Plant**, BEI International, Pergamon Press, 1993.
2. Hambling, P., (Ed.), **Modern Power Station Practice: Turbines, Generators and Associated Plant**, BEI International, Pergamon Press, 1992.
3. Brown, J.G., **Hydroelectric Engineering Practice**, Volumes I, II & III, McGraw-Hill.
4. Hambling, P., (Ed.), **Modern Power Station Practice: Nuclear Turbines, and Associated Plant**, BEI International, Pergamon Press, 1992.
5. Drbal, L. F., Boston, P. G., Westra, K. L., **Black and Veatch, Power Plant Engineering**, Kluwer Academic, 1995.
6. Elliott, T. C., Chen, K., and Swanekamp, R., **Standard Handbook of Power Plant Engineering**, 2nd Edition, McGraw-Hill Professional, 1997.
7. El-Wakil, M. M., **Power Plant Technology**, McGraw-Hill, 1984.
8. Jog, M., **Hydro-electric and Pumped Storage Plants**, John Wiley, 1989.
9. Fritz, J. J., **Small and Mini Hydropower Systems**, McGraw-Hill, 1984.
10. Central Board for Irrigation and Power (CPIB), India, Design and Construction Features of Selected Dams in India, 1983.
11. Borbely, Anne-Marie, and Kreider, Jan J., (Eds.), **Distributed Generation: The Paradigm for the New Millennium**, CRC Press, 2003.
12. Larminie, J., and Dicks, A., **Fuel Cell Systems Explained**, John Wiley, 2003.
13. Vielstich, W., Lamm, A., and Gasteiger, H., **Handbook of Fuel Cells: Fundamentals, Technology, Applications**, John Wiley, 2003.
14. O'Hayre, R., Cha, S. -W., Colella, W., and Prinz, F. B., **Fuel Cell Fundamentals**, Wiley, 2005.
15. Appleby, A. J., and Foulkes, F. R., **Fuel Cell Handbook**, van Nostrand Reinhold, 1996.
16. Harrison, R., Hau, E., and Snel, H., **Large Wind Turbines: Design and Economics**, John Wiley, 2001.
17. Bejan, Adrian, **Advanced Engineering Thermodynamics**, Wiley, 1997.
18. Patents and catalogues related to various equipment.